Periodic Density-Corrected DFT Analytic Gradients Theory and Implementation Conventions

PBC DC-DFT Package Project

Active scope: FFTDF/GDF/RSDF, target multigrid, PBE/SCAN/r2SCAN

Abstract

This document derives the energy and fixed-cell nuclear analytic gradient of density-corrected density functional theory evaluated with semilocal target functionals on restricted closed-shell periodic Hartree–Fock densities. The central complication is that the target energy is not stationary with respect to the HF orbitals. Consequently, the gradient requires the HF CPHF orbital Hessian and an adjoint Z-vector equation. The presentation develops the notation and implementation conventions for FFTDF, Gaussian density fitting (GDF), range-separated density fitting (RSDF), GTH pseudopotentials, Gamma and sampled k-points, and the PBE and SCAN/r2SCAN target functionals in one consistent framework. It also covers a target multigrid integrator selected independently of source GDF/RSDF, matrix-free MINRES Z-vectors, and bounded-memory response algebra. Stress, cell derivatives, multigrid PP, and open-shell response are outside the active scope of this edition.

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1 Purpose, assumptions, and how to read this document

1.1 The physical quantities of interest

Let $\mathbf{R}$ denote the vector collecting all nuclear coordinates. The primary objects studied here are obtained by first solving for the periodic restricted Hartree–Fock (HF) density matrix $D_{\mathrm{HF}}(\mathbf{R})$ and then evaluating a separate semilocal density functional approximation (DFA) on that same density. The resulting energy is

$$E_{\mathrm{DC-DFA}}(\mathbf{R}) = E_{\mathrm{DFA}}[D_{\mathrm{HF}}(\mathbf{R}); \mathbf{R}]\,, \tag{1}$$

and its fixed-cell nuclear gradient is

$$g_{Ax} = \frac{\mathrm{d}E_{\mathrm{DC-DFA}}}{\mathrm{d}R_{Ax}}. \tag{2}$$

Here, A labels an atom and $x \in \{x, y, z\}$ is a Cartesian component. The public API returns the gradient; the force has the opposite sign,

$$F_{Ax} = -g_{Ax}. \tag{3}$$

Equation (1) does not mean "add a DFA correction to the HF energy." The kinetic, electron–nuclear, Hartree, exchange–correlation, and ion–ion terms of the target DFA energy are all re-evaluated on the HF density. The HF energy is retained only as the reference that generated the source orbitals and as a diagnostic quantity.

1.2 Why this differs from an ordinary KS gradient

In a self-consistent KS-DFA calculation, the same DFA energy is stationary with respect to orbital rotations κ. In DC-DFA, the functional that defines the orbitals differs from the functional used to evaluate the energy, so that

$$\left.\frac{\partial E_{\mathrm{DC-DFA}}}{\partial \kappa^{\mathbf{k}}_{ai}}\right|_{D_{\mathrm{HF}}} \neq 0. \tag{4}$$

The change of the HF orbitals under nuclear motion therefore cannot be discarded. Rather than solve a separate CPHF equation for every nuclear displacement, one solves a single adjoint equation for the HF orbital Hessian, the Z-vector equation, and contracts that solution with all perturbations.

1.3 Active scope

This edition treats the following as one complete theoretical and numerical scope:

- restricted closed-shell RHF/KRHF source densities,
- three-dimensional periodic boundary conditions,
- GTH pseudopotentials and atom-centered Gaussian AOs,
- Gamma-point and general sampled k-point meshes,
- PBE, SCAN, and r2SCAN target DFAs,
- FFTDF, GDF, and RSDF two-electron representations,
- GPU target multigrid XC or J+XC selected independently of source DF,
- dense-reference and production matrix-free Z-vectors,
- CPU and GPU implementations, and
- energies and nuclear analytic gradients at fixed lattice vectors.

Stress and cell derivatives, multigrid pseudopotentials, the larger-angular-momentum completion matrix, and open-shell/UHF response are excluded from the active scope and preserved in a separate archive. Accordingly, "analytic gradient" always denotes $\partial E/\partial R_{Ax}$ at fixed cell in this document.

1.4 Prerequisites and organization

The reader is assumed to know AO-based HF and KS-DFT, density matrices, Gaussian integrals, and elementary reciprocal-lattice theory. Section 2 fixes the periodic notation and units, and Sec. 3 defines the target energy evaluated on a fixed HF density. Sections 4 and 6 derive the nonvariational response and the total gradient. Subsequent sections develop the explicit derivatives for GTH, FFTDF, GDF, MGGA, and RSDF, followed by target-multigrid ownership, matrix-free and bounded-memory response algorithms, a worked workflow, and numerical validation procedures.

The word "exact" must be interpreted at two distinct levels. First, within a chosen finite basis, mesh, auxiliary basis, and k-point representation, the analytic derivative must agree with the full-energy finite difference of the same representation. Second, the finite representation itself must converge with cutoff, basis, and k-mesh. The first criterion measures implementation correctness; the second measures physical convergence.

2 Periodic notation, matrix conventions, and units

2.1 Direct and reciprocal lattices

Using the column convention, define the primitive lattice matrix as

$$\mathbf{A} = (\mathbf{a}_1\ \mathbf{a}_2\ \mathbf{a}_3) \tag{5}$$

and the cell volume as

$$\Omega = \det \mathbf{A} > 0. \tag{6}$$

The reciprocal-lattice matrix is

$$\mathbf{B} = 2\pi\mathbf{A}^{-\mathrm{H}}, \qquad \mathbf{a}_i \cdot \mathbf{b}_j = 2\pi\delta_{ij}. \tag{7}$$

A lattice translation is $\mathbf{T} = \mathbf{A}\mathbf{n}$ with $\mathbf{n} \in \mathbb{Z}^3$, and a reciprocal vector is $\mathbf{G} = \mathbf{B}\mathbf{m}$.

Integrals over one cell are denoted by $\int_\Omega \mathrm{d}\mathbf{r}$. Constants in a Fourier pair depend on the FFT normalization used by an implementation; here we adopt

$$\rho(\mathbf{G}) = \int_\Omega e^{-i\mathbf{G}\cdot\mathbf{r}}\rho(\mathbf{r})\mathrm{d}\mathbf{r}, \qquad \rho(\mathbf{r}) = \frac{1}{\Omega}\sum_{\mathbf{G}} e^{i\mathbf{G}\cdot\mathbf{r}}\rho(\mathbf{G}). \tag{8}$$

2.2 Bloch AOs and k-point weights

From an atom-centered Gaussian χ_μ, define a Bloch AO by

$$\phi_{\mu\mathbf{k}}(\mathbf{r}) = \frac{1}{\sqrt{N_\mathrm{c}}}\sum_{\mathbf{T}} e^{i\mathbf{k}\cdot\mathbf{T}}\chi_\mu(\mathbf{r}-\mathbf{T}), \tag{9}$$

where N_c is the number of primitive cells in the Born–von Karman supercell; it cancels from the final intensive quantities. A Bloch orbital is

$$\psi_{p\mathbf{k}}(\mathbf{r}) = \sum_\mu C^{\mathbf{k}}_{\mu p}\phi_{\mu\mathbf{k}}(\mathbf{r}). \tag{10}$$

General weights are normalized as $\sum_{\mathbf{k}} w_{\mathbf{k}} = 1$. Regular meshes in this package use equal weights, $w_{\mathbf{k}} = 1/N_k$. Thus, an explicit factor $1/N_k$ below denotes the equal-weight convention. A Gamma-only calculation is not a separate scalar theory; it is the same tensor convention with $N_k = 1$ and $\mathbf{k} = \mathbf{0}$.

The AO overlap and one-electron matrices are

$$S^{\mathbf{k}}_{\mu\nu} = \langle\phi_{\mu\mathbf{k}}|\phi_{\nu\mathbf{k}}\rangle, \tag{11}$$

$$h^{\mathbf{k}}_{\mu\nu} = \langle\phi_{\mu\mathbf{k}}|\hat{h}|\phi_{\nu\mathbf{k}}\rangle, \tag{12}$$

and the MO coefficients satisfy

$$C^{\mathbf{k}\dagger}S^{\mathbf{k}}C^{\mathbf{k}} = I. \tag{13}$$

2.3 Restricted density matrix

For closed-shell occupations $f_{p\mathbf{k}} \in \{0, 2\}$, the AO density is

$$D^{\mathbf{k}}_{\mu\nu} = \sum_p f_{p\mathbf{k}} C^{\mathbf{k}}_{\mu p} C^{\mathbf{k}*}_{\nu p} = 2\sum_i C^{\mathbf{k}}_{\mu i} C^{\mathbf{k}*}_{\nu i}. \tag{14}$$

This definition already contains the occupation factor of two. Whether an additional factor of two is required in an energy or response contraction therefore depends on whether an occupied-only tensor is being used in place of D.

The matrix-trace convention is

$$\mathrm{Tr}[D^{\mathbf{k}}X^{\mathbf{k}}] = \sum_{\mu\nu} D^{\mathbf{k}}_{\nu\mu}X^{\mathbf{k}}_{\mu\nu}. \tag{15}$$

At complex k-points, $D^{\mathbf{k}}$ and a Hermitian operator are Hermitian, but an individual derivative block need not be. The real part is taken only for the final scalar; intermediate imaginary components must not be discarded arbitrarily.

2.4 Indices and derivative notation

Index	Meaning
$\mu, \nu, \lambda, \sigma$	primary AO
P, Q	auxiliary AO
i, j	occupied MO
a, b	virtual MO
p, q	arbitrary MO
A, B	atom
x, y	Cartesian component
$\mathbf{k}, \mathbf{k}'$	sampled k-point

A total derivative with respect to R_{Ax} is written as a superscript Ax or as ∂_{Ax}; for example, $h^{Ax} = \partial h/\partial R_{Ax}$. Because the AO basis moves with the nuclei, a matrix derivative must be distinguished from the orbital-coefficient response. The cell is fixed, so $\mathbf{A}^{Ax} = 0$, and the reciprocal vectors and discrete mesh are also fixed.

2.5 Canonical units

Quantity	Canonical unit	Comment
Energy	Hartree (Ha)	public energy
Position/lattice	Bohr	PySCF internal unit
Nuclear gradient	Ha/Bohr	$\mathrm{d}E/\mathrm{d}R$
Force	Ha/Bohr	$-\mathrm{d}E/\mathrm{d}R$
k-point	Bohr^{-1}	Cartesian reciprocal coordinate
Plane-wave cutoff	Ha	$G_{\max}^2/2$

Even when an input XYZ file is given in angstrom, the coordinates are in Bohr after the cell is built. Converting a gradient array to eV/angstrom requires both energy and length conversions. Changing the sign is not a unit conversion; it converts a gradient into a force.

3 Target DFA energy evaluated on the HF density

3.1 The source HF problem

The source orbitals are obtained by solving the periodic RHF/KRHF equation self-consistently,

$$F_{\mathrm{HF}}^{\mathbf{k}} C^{\mathbf{k}} = S^{\mathbf{k}} C^{\mathbf{k}} \varepsilon^{\mathbf{k}}. \tag{16}$$

The restricted HF Fock matrix is

$$F_{\mathrm{HF}}^{\mathbf{k}} = h^{\mathbf{k}} + J^{\mathbf{k}}[D] - \frac{1}{2} K^{\mathbf{k}}[D], \tag{17}$$

where D is the density of Eq. (14), including the occupation factor of two. The HF energy per primitive cell is

$$E_{\mathrm{HF}} = \frac{1}{N_k} \sum_{\mathbf{k}} \mathrm{Tr}[D^{\mathbf{k}} h^{\mathbf{k}}] + \frac{1}{2N_k} \sum_{\mathbf{k}} \mathrm{Tr}\left[D^{\mathbf{k}} \left(J^{\mathbf{k}} - \frac{1}{2} K^{\mathbf{k}} \right) \right] + E_{\mathrm{NN}}. \tag{18}$$

FFTDF, GDF, or RSDF determines the numerical representation of J and K in the source HF calculation.

3.2 Target energy at a frozen HF density

After the source HF calculation has converged, a target RKS/KRKS object is constructed and the target functional is evaluated while the density remains fixed. For a semilocal target DFA,

$$E_{\mathrm{DC-DFA}}[D] = E_1[D] + E_{\mathrm{H}}[D] + E_{\mathrm{xc}}^{\mathrm{DFA}}[D] + E_{\mathrm{NN}}, \tag{19}$$

with

$$E_1[D] = \frac{1}{N_k} \sum_{\mathbf{k}} \mathrm{Tr}[D^{\mathbf{k}} h^{\mathbf{k}}], \tag{20}$$

$$E_{\mathrm{H}}[D] = \frac{1}{2N_k} \sum_{\mathbf{k}} \mathrm{Tr}[D^{\mathbf{k}} J^{\mathbf{k}}[D]]. \tag{21}$$

PBE, SCAN, and r2SCAN are pure semilocal target functionals, so the target energy contains no exact-exchange matrix K. Nevertheless, because D is an HF solution, source HF exchange remains in the orbital Hessian that determines the response.

The target Fock matrix, namely the density derivative of $E_{\mathrm{DC-DFA}}$, is

$$F_{\mathrm{t}}^{\mathbf{k}} = h^{\mathbf{k}} + J^{\mathbf{k}}[D] + V_{\mathrm{xc}}^{\mathbf{k}}[D]. \tag{22}$$

The subscript t denotes "target." Once D_{HF} has been passed to the target object, no KS orbital optimization is performed.

3.3 Density and semilocal variables

The real-space density is

$$\rho(\mathbf{r}) = \frac{1}{N_k} \sum_{\mathbf{k}} \sum_{\mu\nu} D_{\mu\nu}^{\mathbf{k}} \phi_{\nu\mathbf{k}}^*(\mathbf{r}) \phi_{\mu\mathbf{k}}(\mathbf{r}). \tag{23}$$

A spin-unpolarized GGA may be written as

$$E_{\mathrm{xc}}^{\mathrm{GGA}} = \int_\Omega f(\rho, \sigma)\, \mathrm{d}\mathbf{r}, \qquad \sigma = |\nabla\rho|^2. \tag{24}$$

SCAN and r2SCAN also depend on the positive kinetic-energy density

$$\tau(\mathbf{r}) = \frac{1}{2N_k} \sum_{\mathbf{k}} \sum_{\mu\nu} D_{\mu\nu}^{\mathbf{k}} \nabla\phi_{\nu\mathbf{k}}^*(\mathbf{r}) \cdot \nabla\phi_{\mu\mathbf{k}}(\mathbf{r}), \tag{25}$$

so that

$$E_{\mathrm{xc}}^{\mathrm{MGGA}} = \int_\Omega f(\rho, \sigma, \tau) \mathrm{d}\mathbf{r}. \tag{26}$$

The PySCF SCAN/r2SCAN route does not use a density-Laplacian channel. The factor 1/2 in Eq. (25) is part of the convention and must be included exactly once in the contraction with $v_\tau = \partial f / \partial \tau$.

3.4 Nonvariational orbital gradient

Apply an occupied–virtual anti-Hermitian rotation to the MO coefficients,

$$C^{\mathbf{k}} \longrightarrow C^{\mathbf{k}} e^{\kappa^{\mathbf{k}}}, \qquad \kappa^{\mathbf{k}\dagger} = -\kappa^{\mathbf{k}}. \tag{27}$$

Including the conventional closed-shell factor of two, the first variation of the energy may be written as

$$\delta E_{\mathrm{DC-DFA}} = \frac{2}{N_k} \mathrm{Re} \sum_{\mathbf{k}ai} F_{\mathrm{t},ai}^{\mathbf{k}}\, \delta\kappa_{ai}^{\mathbf{k}}. \tag{28}$$

Canonical source HF orbitals satisfy $F_{\mathrm{HF},ai}^{\mathbf{k}} = 0$, but in general

$$F_{\mathrm{t},ai}^{\mathbf{k}} \neq 0. \tag{29}$$

This nonzero target virtual–occupied block is the right-hand side of the Z-vector equation.

3.5 Metadata required by the energy object

Even for the same named DFA, different bases, pseudopotentials, cells, k-points, DF backends, and numerical grids define different finite energies. A reproducible energy object must record at least the target functional, source reference, N_k, lattice and coordinate units, primary and auxiliary bases, GTH pseudopotential, DF backend, and either the FFT mesh or the RSDF source and metric parameters. The gradient must differentiate precisely this same energy object and its resolved settings.

4 Periodic orbital response and the Z-vector

4.1 Response of the orthonormality constraint

Under a nuclear displacement R_{Ax}, the AO overlap changes, so the MO coefficient derivative is not merely a unitary rotation. Differentiating Eq. (13) gives

$$C^{\mathbf{k}\dagger} S^{\mathbf{k}} C^{\mathbf{k},Ax} + C^{\mathbf{k},Ax\dagger} S^{\mathbf{k}} C^{\mathbf{k}} + C^{\mathbf{k}\dagger} S^{\mathbf{k},Ax} C^{\mathbf{k}} = 0. \tag{30}$$

Defining $U^{\mathbf{k},Ax} = C^{\mathbf{k}\dagger} S^{\mathbf{k}} C^{\mathbf{k},Ax}$ yields

$$U^{\mathbf{k},Ax} + U^{\mathbf{k},Ax\dagger} = -S^{\mathbf{k},Ax}_{\mathrm{MO}}, \qquad S^{\mathbf{k},Ax}_{\mathrm{MO}} = C^{\mathbf{k}\dagger} S^{\mathbf{k},Ax} C^{\mathbf{k}}. \tag{31}$$

The occupied–occupied and virtual–virtual gauges may be chosen, whereas the occupied–virtual block is determined by the HF response equation.

4.2 The CPHF equation

Differentiating the HF stationarity condition $F^{\mathbf{k}}_{ai} = 0$ with respect to a nuclear coordinate gives

$$\sum_{\mathbf{k}'bj} A_{ai\mathbf{k},bj\mathbf{k}'} U^{\mathbf{k}',Ax}_{bj} = -B^{\mathbf{k},Ax}_{ai}. \tag{32}$$

Here A is the periodic restricted HF orbital Hessian and B^{Ax} is the nuclear perturbation source. In canonical-orbital notation, the Hessian action has the schematic form

$$(AU)^{\mathbf{k}}_{ai} = (\varepsilon^{\mathbf{k}}_a - \varepsilon^{\mathbf{k}}_i) U^{\mathbf{k}}_{ai} + \left[C^{\mathbf{k}\dagger}_v \, \delta G_{\mathrm{HF}}[D^{(1)}(U)]^{\mathbf{k}} C^{\mathbf{k}}_o \right]_{ai}. \tag{33}$$

Since $\delta G_{\mathrm{HF}} = \delta J - \frac{1}{2}\delta K$, applying the selected FFTDF/GDF/RSDF J/K provider to a transition density is the core AX operation.

The nuclear source contains one-electron, two-electron, and overlap terms,

$$B^{\mathbf{k},Ax}_{ai} = \left[C^{\mathbf{k}\dagger}_v \left(h^{\mathbf{k},Ax} + G^{\mathbf{k},Ax}_{\mathrm{HF}} - \varepsilon^{\mathbf{k}}_i S^{\mathbf{k},Ax} \right) C^{\mathbf{k}}_o \right]_{ai} + B^{\mathbf{k},Ax}_{ai,\mathrm{basis}}. \tag{34}$$

G^{Ax}_{HF} is the integral derivative at fixed density. The final term may be redistributed among other terms depending on the chosen AO-row or AO-column convention; production code and its documentation must use the same convention.

4.3 The adjoint Z-vector

Solving Eq. (32) independently for all $3N_{\mathrm{atom}}$ displacements is expensive. Collect the target orbital gradient into a vector g and solve the adjoint equation once,

$$A^{\mathrm{H}} z = -g. \tag{35}$$

The chain-rule contribution then becomes

$$g^{\mathrm{resp}}_{Ax} = z^{\mathrm{H}} B^{Ax}. \tag{36}$$

The signs in Eqs. (35) and (36) follow from defining the CPHF equation as $AU = -B$. If the definition of B or g changes, both signs must be changed consistently.

4.4 The real-linear problem at complex k-points

For a complex orbital rotation $U = X + iY$, the Hermitian first-order density is

$$D^{(1),\mathbf{k}}(U) = 2\left(C_v^{\mathbf{k}} U^{\mathbf{k}} C_o^{\mathbf{k}\dagger} + C_o^{\mathbf{k}} U^{\mathbf{k}\dagger} C_v^{\mathbf{k}\dagger}\right). \tag{37}$$

Because U^* also appears, the map $U \mapsto D^{(1)}$ is real-linear rather than complex-linear. The solver coordinate is therefore realified as

$$u_R = \begin{pmatrix} \mathrm{Re}\, U \\ \mathrm{Im}\, U \end{pmatrix}, \tag{38}$$

and the equations solved are

$$A_R u_R = -B_R, \qquad A_R^T z_R = -g_R. \tag{39}$$

The fact that Gamma-point matrices can be real is merely a simplifying special case of this general formulation.

4.5 k-point normalization and momentum structure

A nuclear perturbation preserves primitive-cell periodicity and therefore generates a VO response within each $\mathbf{k}$. The J/K Hessian action, however, couples density responses from all k-points. Under the equal-weight convention, both AX and the final scalar contraction must contain exactly one factor $1/N_k$. Adding another factor of two without checking whether the transition density already contains the occupation factor may accidentally work at Gamma while breaking sampled-k consistency.

4.6 Gauge freedom, small gaps, and residuals

Unitary rotations within the occupied or virtual subspaces leave the density unchanged. For degenerate orbitals, the subspace-invariant density response, not an individual U_{ai}, is the physical quantity. The solver operates only in the VO space and must not silently regularize small denominators. At a minimum, the report records

$$r_{\max} = \|A_R^T z_R + g_R\|_\infty \tag{40}$$

together with the response dimension and a condition estimate. Even good gradient–FD agreement is insufficient evidence if $r_{\max}$ is large, because the result may depend on cancellation for a particular perturbation.

5 Periodic density fitting and RSDF response

5.1 Why density-fitting derivatives are required

Direct treatment of four-center electron-repulsion integrals causes the cost and memory of periodic HF and target-DFA Coulomb and exact-exchange energies to grow rapidly. Density fitting expands AO-pair densities in an auxiliary Gaussian basis and replaces the four-center problem by three-center quantities and a two-center metric. When the nuclei move, however, both the primary AOs and the auxiliary basis move with them. Thus, if density fitting is used for the energy value, the gradient must include the auxiliary-basis and metric responses.

Denote a Bloch AO by $\chi_{\mu\mathbf{k}}(\mathbf{r})$ and define the momentum transfer as

$$\mathbf{q} = \mathbf{k}_j - \mathbf{k}_i. \tag{41}$$

For an auxiliary Bloch function $P\mathbf{q}$, the three-center source and the Coulomb metric are

$$B^{ij}_{\mu\nu,P} = \left(\chi_{\mu\mathbf{k}_i}\chi_{\nu\mathbf{k}_j}\middle|P\mathbf{q}\right), \tag{42}$$

$$M_{PQ}(\mathbf{q}) = (P\mathbf{q}|Q\mathbf{q}). \tag{43}$$

Whether the ket AO is represented with an explicit complex conjugate depends on the integral-orientation convention of the program. The essential requirement is that the central value and its derivative use the same convention.

For a positive-definite metric, let

$$\mathbf{M}(\mathbf{q}) = \mathbf{L}(\mathbf{q})\mathbf{L}(\mathbf{q})^\dagger \tag{44}$$

be its Cholesky decomposition. The orthonormalized three-center factor is

$$\mathbf{C}^{ij} = \mathbf{L}(\mathbf{q})^{-1}\mathbf{B}^{ij}. \tag{45}$$

A four-center integral is then approximated by a factorization of the form

$$(\mu i, \nu j|\lambda l, \sigma m) \simeq \sum_P C^{ij}_{P,\mu\nu} C^{lm*}_{P,\lambda\sigma}. \tag{46}$$

The factor $\mathbf{C}$ corresponds to PySCF's CDERI.

5.2 Nuclear derivatives and auxiliary response

Differentiating Eq. (45) with respect to a Cartesian coordinate R_{Ax} gives

$$\mathbf{C}^{Ax} = \mathbf{L}^{-1}\mathbf{B}^{Ax} - \mathbf{L}^{-1}\mathbf{L}^{Ax}\mathbf{C}. \tag{47}$$

The first term is the three-center source response and the second is the metric response. Freezing the metric omits the motion of the auxiliary Gaussian centers and the change of the fitting coefficients. This omission may be accidentally small for a small numerator, but it generally grows as the number of auxiliary centers increases.

The Cholesky derivative satisfies

$$\mathbf{M}^{Ax} = \mathbf{L}^{Ax}\mathbf{L}^\dagger + \mathbf{L}\mathbf{L}^{Ax\dagger}. \tag{48}$$

Multiplying by $\mathbf{L}^{-1}$ and $\mathbf{L}^{-\dagger}$ defines

$$\mathbf{W} = \mathbf{L}^{-1}\mathbf{M}^{Ax}\mathbf{L}^{-\dagger}. \tag{49}$$

In the lower-Cholesky gauge,

$$\mathbf{L}^{Ax} = \mathbf{L}\,\Phi(\mathbf{W}), \tag{50}$$

where Φ preserves the strict lower triangle and takes half of the Hermitian diagonal. The same gauge must be used consistently for the central value and the derivative.

5.3 Range separation of the Coulomb kernel

RSDF partitions the Coulomb kernel according to

$$\frac{1}{r} = \frac{\mathrm{erfc}(\omega r)}{r} + \frac{\mathrm{erf}(\omega r)}{r}. \tag{51}$$

The first term decays rapidly in real space and is evaluated as a lattice sum; the second is evaluated as a smooth reciprocal-space Fourier sum. This is an exact identity, but with finite

image lists and reciprocal meshes the value of ω, the screening precision, and the mesh become part of the numerical representation.

In the public PySCF RSDF implementation, optimal parameters for the three-center source and the two-center metric may be determined independently. Explicitly,

$$\mathbf{B}^{ij} = \mathbf{B}^{ij}_{\mathrm{SR}}(\omega_B, \epsilon_R) + \mathbf{B}^{ij}_{\mathrm{LR}}(\omega_B, \mathcal{G}_B) + \delta_{\mathbf{q},0}\mathbf{B}^{ij}_{vbar}, \tag{52}$$

$$\mathbf{M}(\mathbf{q}) = \mathbf{M}_{\mathrm{SR}}(\mathbf{q}; \omega_M, \epsilon_M) + \mathbf{M}_{\mathrm{LR}}(\mathbf{q}; \omega_M, \mathcal{G}_M) + \delta_{\mathbf{q},0}\mathbf{M}_{G=0}. \tag{53}$$

In general, $\omega_B \neq \omega_M$ and $\mathcal{G}_B \neq \mathcal{G}_M$. The exponent distribution of the auxiliary basis differs from that of the primary AO pairs, and the metric may therefore require a higher cutoff.

At fixed cell, lattice and reciprocal vectors do not change under nuclear motion. Holding ω and the mesh fixed as numerical inputs gives

$$\mathbf{B}^{ij,Ax} = \mathbf{B}^{ij,Ax}_{\mathrm{SR}} + \mathbf{B}^{ij,Ax}_{\mathrm{LR}} + \delta_{\mathbf{q},0}\mathbf{B}^{ij,Ax}_{vbar}, \tag{54}$$

$$\mathbf{M}^{Ax} = \mathbf{M}^{Ax}_{\mathrm{SR}} + \mathbf{M}^{Ax}_{\mathrm{LR}} + \delta_{\mathbf{q},0}\mathbf{M}^{Ax}_{G=0}. \tag{55}$$

For stress, the lattice and reciprocal mesh themselves vary; the fixed-cell formulas in this section are therefore not sufficient for a cell derivative.

5.4 Ownership of primary-AO and auxiliary centers

A three-center integral contains three atom-centered objects. Symbolically,

$$B^{Ax}_{\mu\nu,P} = \delta_{A,A_\mu}\partial_{A_\mu x}B_{\mu\nu,P} + \delta_{A,A_\nu}\partial_{A_\nu x}B_{\mu\nu,P} + \delta_{A,A_P}\partial_{A_P x}B_{\mu\nu,P}. \tag{56}$$

Integral-library operators such as `ip1` and `ip2` often return derivatives with respect to an electronic coordinate. Translational identities then introduce the opposite sign for a derivative that moves a nuclear center. The three contributions must be accumulated after applying the atom slices for the AO row, AO column, and auxiliary index.

The two-center metric has two auxiliary centers,

$$M^{Ax}_{PQ} = \delta_{A,A_P}\partial_{A_P x}M_{PQ} + \delta_{A,A_Q}\partial_{A_Q x}M_{PQ}. \tag{57}$$

The short-range metric derivative must use the same screened lattice-image list as the central value. Different image cutoffs may both approximate the continuous Coulomb metric, but they do not define an exact derivative of the same finite calculation.

5.5 The $G = 0$ mode and charge corrections

The $G = 0$ mode of the periodic Coulomb kernel is treated specially according to charge neutrality and the Ewald convention. The source `vbar` and the metric $G = 0$ correction occur only in blocks with exactly zero momentum transfer,

$$\delta_{\mathbf{q},0} = \begin{cases} 1, & \mathbf{k}_j - \mathbf{k}_i = \mathbf{0}, \\ 0, & \text{otherwise.} \end{cases} \tag{58}$$

This includes not only Gamma-only calculations but also diagonal pairs in a sampled-k calculation. Therefore, “is this Gamma?” and “is the momentum transfer zero?” are not equivalent tests.

5.6 k-point pairs and Hermitian reconstruction

For N_k k-points, the J/K build requires multiple (i, j) pairs. Momentum conservation groups the metric by $\mathbf{q} = \mathbf{k}_j - \mathbf{k}_i$. If only canonical pairs are stored, a reverse pair is reconstructed through a Hermitian relation such as

$$C^{ji}_{P,\nu\mu} = C^{ij*}_{P,\mu\nu}. \tag{59}$$

Its nuclear derivative must obey the same conjugation and AO-index transpose. At complex k-points, all pair contractions and the equal-weight $1/N_k$ normalization must be consistent for the final gradient to be real.

Define the closed-shell Coulomb auxiliary density by

$$j_P = \sum_{\mathbf{k}\mu\nu} C^{\mathbf{k}}_{P,\mu\nu} D^{\mathbf{k}}_{\nu\mu}. \tag{60}$$

Its fixed-density J derivative is

$$(J^{\mathbf{k}})^{Ax}_{\mu\nu} = \frac{1}{N_k} \sum_P \left[j^{Ax}_P C^{\mathbf{k}}_{P,\mu\nu} + j_P C^{\mathbf{k},Ax}_{P,\mu\nu} \right]. \tag{61}$$

Exchange contains two CDERI factors, so the product rule must be applied to each factor.

5.7 Role within the DC-DFT gradient

RSDF determines the numerical representation of the two-electron value and response; the nonvariational structure of DC-DFT itself is independent of the backend. The total gradient may be written as

$$E^{Ax}_{\mathrm{DC}} = E^{Ax}_{\mathrm{target,explicit}} + \mathbf{z}^T \mathbf{B}^{Ax}_{\mathrm{HF}}. \tag{62}$$

The first term contains the target one-electron, Hartree, XC, overlap/Pulay, and nuclear contributions. The second includes the orbital response of the HF density to nuclear motion through the adjoint Z-vector. RSDF C^{Ax} enters both the explicit target J derivative and the source HF response operator.

5.8 Validation principles

Correctness of the analytic derivative and physical convergence are separate questions. First compare with a central finite difference of the same finite RSDF representation,

$$g^{\mathrm{FD}}_{Ax}(h) = \frac{E(\mathbf{R} + h\mathbf{e}_{Ax}) - E(\mathbf{R} - h\mathbf{e}_{Ax})}{2h}. \tag{63}$$

Then converge h, the source and metric meshes, the primary and auxiliary bases, and the k-mesh independently.

In addition, a separately reconstructed CDERI must agree with the SCF storage,

$$\Delta_C = \max_{ijP\mu\nu} \left| C^{ij,\mathrm{reconstructed}}_{P,\mu\nu} - C^{ij,\mathrm{storage}}_{P,\mu\nu} \right|. \tag{64}$$

If Δ_C is not small, even a mathematically correct derivative formula is differentiating a representation with a different ω, mesh, pair orientation, or factor gauge. Such a derivative must be rejected for production use.

6 The DC-DFT Lagrangian and the total nuclear gradient

6.1 A Lagrangian with the HF constraint

Let C denote the HF orbital coefficients and define the VO block of the HF stationarity residual as

$$f_{ai}^{\mathbf{k}}(C,\mathbf{R}) = \left[C_v^{\mathbf{k}\dagger} F_{\mathrm{HF}}^{\mathbf{k}} C_o^{\mathbf{k}}\right]_{ai}. \tag{65}$$

For a converged HF solution, $f = 0$. The target energy is not stationary under this constraint, so introduce an adjoint multiplier z and define the reduced Lagrangian

$$\mathcal{L}(C,z;\mathbf{R}) = E_{\mathrm{DC-DFA}}(C;\mathbf{R}) + 2\,\mathrm{Re}\sum_{\mathbf{k}ai} w_{\mathbf{k}} z_{ai}^{\mathbf{k}*} f_{ai}^{\mathbf{k}}(C;\mathbf{R}). \tag{66}$$

The multiplier enforcing MO orthonormality is represented below through the energy-weighted density and the overlap/Pulay term.

Stationarity with respect to orbital rotations,

$$\frac{\partial \mathcal{L}}{\partial \kappa} = 0, \tag{67}$$

gives the Z-vector equation of Eq. (35). Explicit $\mathrm{d}C/\mathrm{d}R_{Ax}$ terms then disappear from the total derivative, leaving

$$\frac{\mathrm{d}E_{\mathrm{DC-DFA}}}{\mathrm{d}R_{Ax}} = \left.\frac{\partial \mathcal{L}}{\partial R_{Ax}}\right|_{C,z} = g_{Ax}^{\mathrm{explicit}} + g_{Ax}^{\mathrm{response}}. \tag{68}$$

6.2 Explicit target contribution

The explicit derivative of the target energy at fixed density is decomposed as

$$g_{Ax}^{\mathrm{explicit}} = g_{Ax}^{1e} + g_{Ax}^{\mathrm{H}} + g_{Ax}^{\mathrm{xc}} + g_{Ax}^{\mathrm{Pulay,t}} + g_{Ax}^{\mathrm{NN}}. \tag{69}$$

The one-electron term is

$$g_{Ax}^{1e} = \frac{1}{N_k}\sum_{\mathbf{k}} \mathrm{Tr}[D^{\mathbf{k}} h^{\mathbf{k},Ax}], \tag{70}$$

where h^{Ax} contains the AO-matrix derivatives of the kinetic, local GTH, and nonlocal GTH operators.

At fixed density, the Hartree term is

$$g_{Ax}^{\mathrm{H}} = \frac{1}{2N_k}\sum_{\mathbf{k}} \mathrm{Tr}[D^{\mathbf{k}} J^{\mathbf{k},Ax}[D]]. \tag{71}$$

For FFTDF, $J^{Ax}[D]$ contains the AO/grid derivative and the appropriate Coulomb-kernel convention. For GDF and RSDF it contains the three-center source and two-center metric derivatives.

The explicit semilocal XC derivative differentiates the AOs and the quadrature representation while holding the density matrix fixed,

$$g_{Ax}^{\mathrm{xc}} = \left.\frac{\partial E_{\mathrm{xc}}[D;\mathbf{R}]}{\partial R_{Ax}}\right|_{D}. \tag{72}$$

Atom-centered Gaussian AOs are evaluated on a uniform periodic grid, so the AO center derivative and periodic-image/Bloch-phase conventions must match those of the central value. SCAN and r2SCAN add the τ channel.

6.3 The target overlap/Pulay term

Define the target energy-weighted density from the occupied–occupied block of the target Fock matrix,

$$W_{\mathrm{t}}^{\mathbf{k}} = 2C_o^{\mathbf{k}} \left(C_o^{\mathbf{k}\dagger} F_{\mathrm{t}}^{\mathbf{k}} C_o^{\mathbf{k}} \right) C_o^{\mathbf{k}\dagger}. \tag{73}$$

The target overlap contribution is then

$$g_{Ax}^{\mathrm{Pulay,t}} = -\frac{1}{N_k} \sum_{\mathbf{k}} \mathrm{Tr}[W_{\mathrm{t}}^{\mathbf{k}} S^{\mathbf{k},Ax}]. \tag{74}$$

An AO-row derivative formulation may redistribute parts of Eqs. (70)–(74) among atom-row contractions, but their sum must be identical.

Using GTH ionic charges and the periodic Ewald convention, the ion–ion term is

$$g_{Ax}^{\mathrm{NN}} = \frac{\partial E_{\mathrm{NN}}}{\partial R_{Ax}}. \tag{75}$$

6.4 The HF orbital-response contribution

Writing Eq. (36) explicitly in indices gives

$$g_{Ax}^{\mathrm{response}} = 2\,\mathrm{Re} \sum_{\mathbf{k}ai} w_{\mathbf{k}} z_{ai}^{\mathbf{k}*} B_{ai}^{\mathbf{k},Ax}. \tag{76}$$

Here B^{Ax} is the nuclear derivative of the source HF equation. Changing the target between PBE and an MGGA does not change the physical definitions of B^{Ax} or the HF Hessian. Only the target VO gradient g and the explicit XC derivative change, so the same response infrastructure can be reused.

6.5 Backend-independent assembly

The final package decomposition is recorded as

$$g_{Ax}^{\mathrm{elec,explicit}} = g_{Ax}^{1e} + g_{Ax}^{\mathrm{H}} + g_{Ax}^{\mathrm{xc}} + g_{Ax}^{\mathrm{Pulay,t}}, \tag{77}$$

$$g_{Ax} = g_{Ax}^{\mathrm{elec,explicit}} + g_{Ax}^{\mathrm{NN}} + g_{Ax}^{\mathrm{response}}. \tag{78}$$

FFTDF, GDF, and RSDF differ in their providers for J/K values and derivatives; the Z-vector equation and final assembly contract are shared.

Note 6.1. A full-energy finite difference reconverges the source HF solution and re-evaluates the target energy at both $\mathbf{R} + h\mathbf{e}_{Ax}$ and $\mathbf{R} - h\mathbf{e}_{Ax}$. A component finite difference with the central density held fixed is useful for diagnosing an individual kernel such as Eq. (72), but it is not a reference for the total analytic gradient.

6.6 Translational invariance

Translating all atoms by the same vector within a fixed lattice must not change the energy. In a converged representation,

$$\sum_A \mathbf{g}_A = \mathbf{0}. \tag{79}$$

A large net gradient is a sensitive diagnostic for missing AO-image derivatives, auxiliary-center response, XC-grid aliasing, $G = 0$ treatment, or k-point normalization.

7 One-electron operators and GTH pseudopotential derivatives

7.1 The periodic one-electron Hamiltonian

With a GTH pseudopotential, the one-electron Hamiltonian is

$$\hat{h} = -\frac{1}{2}\nabla^2 + \sum_A \hat{V}^A_{\rm loc} + \sum_A \hat{V}^A_{\rm nl}. \tag{80}$$

The all-electron electron–nuclear potential $-Z_A/r$ is replaced by a smooth local potential and a finite-rank nonlocal projector. Although the kinetic operator has no explicit nuclear-coordinate dependence, its AO matrix derivative is nonzero because the atom-centered basis moves.

7.2 The local GTH potential

A representative local form is

$$V^A_{\rm loc}(r) = -\frac{Z^{\rm ion}_A}{r}\,\mathrm{erf}\left(\frac{r}{\sqrt{2}r_{{\rm loc},A}}\right) + e^{-r^2/(2r^2_{{\rm loc},A})}\sum_{n=1}^{4} C_{nA}\left(\frac{r}{r_{{\rm loc},A}}\right)^{2n-2}. \tag{81}$$

In a periodic FFT representation, the position dependence is carried by the structure factor $e^{-i\mathbf{G}\cdot\mathbf{R}_A}$, and therefore

$$\frac{\partial}{\partial R_{Ax}}\left[e^{-i\mathbf{G}\cdot\mathbf{R}_A}V^A_{\rm loc}(\mathbf{G})\right] = -iG_x e^{-i\mathbf{G}\cdot\mathbf{R}_A}V^A_{\rm loc}(\mathbf{G}). \tag{82}$$

At $\mathbf{G} = 0$, $G_x = 0$, but the constant term, charge convention, and Ewald compensation must still match those used by the central hcore.

The AO matrix is

$$V^{\mathbf{k}}_{\mu\nu,{\rm loc}} = \langle\phi_{\mu\mathbf{k}}|\hat{V}_{\rm loc}|\phi_{\nu\mathbf{k}}\rangle. \tag{83}$$

Its derivative contains the motion of potential center A, the bra AO center, and the ket AO center. Translational identities can express the potential-center derivative through a combination of AO derivatives and the reciprocal-space structure-factor derivative.

7.3 Nonlocal projectors

The nonlocal GTH operator has a separable form for each angular-momentum channel,

$$\hat{V}^A_{\rm nl} = \sum_{lm}\sum_{ij}|p^A_{i,lm}\rangle h^{A,l}_{ij}\langle p^A_{j,lm}|. \tag{84}$$

Using the projector overlap

$$P^{A\mathbf{k}}_{i,lm;\mu} = \langle p^A_{i,lm}|\phi_{\mu\mathbf{k}}\rangle, \tag{85}$$

the AO matrix element is

$$V^{\mathbf{k}}_{\mu\nu,{\rm nl}} = \sum_{A,lm,ij} P^{A\mathbf{k}*}_{i,lm;\mu}h^{A,l}_{ij}P^{A\mathbf{k}}_{j,lm;\nu}. \tag{86}$$

The product rule gives

$$V^{\mathbf{k},Ax}_{\mu\nu,{\rm nl}} = \sum_{lmij}\Big[P^{A\mathbf{k},Ax*}_{i,lm;\mu}h^{A,l}_{ij}P^{A\mathbf{k}}_{j,lm;\nu} + P^{A\mathbf{k}*}_{i,lm;\mu}h^{A,l}_{ij}P^{A\mathbf{k},Ax}_{j,lm;\nu}\Big] + \text{AO-center terms}. \tag{87}$$

The radial projector table $h^{A,l}_{ij}$ moves rigidly with the atom and has no parameter derivative with respect to the nuclear coordinate.

7.4 AO-row conventions and the full matrix

An integral engine often returns a derivative matrix $h^{(1)}_{\mu\nu}$ with respect to the AO-row center. The full matrix derivative for atom A must combine bra, ket, and operator-center ownership,

$$h^{Ax} = h^{x}_{\mathrm{row},A} + h^{x\dagger}_{\mathrm{row},A} + h^{x}_{\mathrm{operator},A}. \tag{88}$$

At complex k-points, the second term is a conjugate transpose, not merely a transpose.

The overlap derivative obeys the analogous AO-center identity

$$S^{Ax} = S^{x}_{\mathrm{row},A} + S^{x\dagger}_{\mathrm{row},A}. \tag{89}$$

Both h^{Ax} and S^{Ax} enter the explicit target gradient and the source HF CPHF source in Eq. (34).

7.5 Pseudopotential consistency

Using an HF source density does not create a distinct analytic form called an "HF pseudopotential." A label such as GTH-HF identifies a pseudopotential parameter set. The source HF calculation and the target-DFA energy and gradient must use the same GTH parameters and valence ionic charges. Using different pseudopotentials for source and target would no longer correspond to differentiating a single energy surface of Eq. (1).

Validation first compares local, nonlocal, kinetic/overlap, and ion–ion components separately with finite differences and then checks the full hcore derivative. Atom-slice and conjugation errors become especially visible when the nonlocal projector contains more angular channels.

8 FFTDF: reciprocal-space Coulomb and exchange derivatives

8.1 Uniform-grid and plane-wave representations

FFTDF evaluates AO-pair densities on a uniform real-space grid and transforms them to reciprocal space with an FFT. Schematically, the total-density Fourier coefficient generated by the density matrices is

$$\rho_D(\mathbf{G}) = \frac{1}{N_k} \sum_{\mathbf{k}\mu\nu} D^{\mathbf{k}}_{\mu\nu} \int_\Omega e^{-i\mathbf{G}\cdot\mathbf{r}} \phi^*_{\nu\mathbf{k}}(\mathbf{r}) \phi_{\mu\mathbf{k}}(\mathbf{r}) \mathrm{d}\mathbf{r}. \tag{90}$$

For a neutral periodic system, excluding the $\mathbf{G} = 0$ component, the Hartree energy is

$$E_{\mathrm{H}} = \frac{1}{2\Omega} \sum_{\mathbf{G}\neq 0} \frac{4\pi}{G^2} |\rho_D(\mathbf{G})|^2. \tag{91}$$

Inverse-transforming the Hartree potential and integrating it with the AO pairs produces $J^{\mathbf{k}}$.

8.2 Nuclear derivatives at fixed mesh

At fixed cell and fixed integer mesh, neither the grid points nor the $\mathbf{G}$ set change under a nuclear displacement. The nuclear derivative of the Coulomb kernel $4\pi/G^2$ therefore vanishes, leaving only the derivative of the AO-pair density,

$$E^{Ax}_{\mathrm{H}} = \frac{1}{\Omega} \mathrm{Re} \sum_{\mathbf{G}\neq 0} \frac{4\pi}{G^2} \rho_D(\mathbf{G})^* \rho^{Ax}_D(\mathbf{G}). \tag{92}$$

ρ^{Ax}_D contains the AO row and column derivatives for atom A and the same periodic-image and Bloch-phase conventions as the central density. Constructing an AO-row matrix followed by Hermitian reconstruction yields the full J^{Ax} entering Eq. (71).

8.3 Exact exchange and momentum transfer

Source HF exchange uses the Coulomb kernel $4\pi/|\mathbf{G}+\mathbf{q}|^2$ for each pair momentum transfer

$$\mathbf{q} = \mathbf{k}' - \mathbf{k}. \tag{93}$$

Symbolically,

$$K^{\mathbf{k}}_{\mu\nu} = \frac{1}{N_k}\sum_{\mathbf{k}'}\sum_{\lambda\sigma}(\mu\mathbf{k}, \lambda\mathbf{k}'|\sigma\mathbf{k}', \nu\mathbf{k})D^{\mathbf{k}'}_{\lambda\sigma}. \tag{94}$$

The nuclear derivative applies the product rule to whichever of the four AO centers moves. The same FFT exchange provider must be applied to transition densities in the source HF Hessian action.

8.4 $G = 0$, $q = 0$, and the Ewald exchange divergence

The Hartree $\mathbf{G} = 0$ mode is omitted or corrected consistently with the neutralizing convention and GTH ionic charges. Exchange is singular at $\mathbf{q} = 0$, $\mathbf{G} = 0$, and PySCF's `exxdiv='ewald'` convention adds an overlap-dependent correction. Because the overlap changes under nuclear motion, this correction also requires a derivative of the form

$$K^{Ax}_{\text{Ewald}} \propto S^{Ax}DS + SDS^{Ax}. \tag{95}$$

Confusing "Gamma" with "$\mathbf{q} = 0$" loses the singular correction for the diagonal pairs of a sampled-k calculation.

8.5 Cutoff and mesh

The plane-wave kinetic cutoff is approximately

$$E_{\text{cut}} = \frac{G^2_{\max}}{2}. \tag{96}$$

Considering the Nyquist wavevector along each direction of an orthorhombic cell, the mesh size scales with the cell length and with $\sqrt{E_{\text{cut}}}$. PySCF couples `cell.ke_cutoff` and `cell.mesh`, but integer rounding means that a small cutoff change can change the mesh discontinuously.

Both $\mathbf{R} + h$ and $\mathbf{R} - h$ in a full-energy finite difference must use the same lattice and the same integer mesh. Re-estimating the mesh from the cutoff at each displaced geometry mixes a change of numerical representation into the finite difference. Metadata must therefore record both the cutoff and the resolved mesh.

8.6 Terms owned by FFTDF in the gradient

A consistent FFTDF provider supplies

1. the central source-HF $J[D]$ and $K[D]$,
2. δJ and δK for transition densities in AX,
3. the full fixed-density $J^{Ax}[D]$ and $K^{Ax}[D]$,
4. the $\mathbf{q} = 0$ Ewald exchange-divergence derivative, and
5. the explicit target-Hartree derivative.

Target XC is owned by the NumInt/MGGA grid provider, while the GTH one-electron derivative is owned by a separate one-electron provider. Keeping these responsibilities separate avoids component double counting even when the same mesh is used.

8.7 CPU and GPU ownership

The CPU route uses the PySCF periodic FFTDF central value and project-local derivative matrices. The GPU route uses GPU4PySCF FFTs, AO grids, and CuPy contractions. Cell and lattice objects, small pseudopotential tables, and final reporting may remain on the host, but AO/grid FFTs, J/K, AX, the Z-vector, and final electronic assembly must be device operations. The capability report discloses host boundaries and forbids silent CPU fallback.

FFTDF validation does not end when a finite difference agrees at one E_{cut}. Energy and gradient must converge along a resolved mesh sequence, Gamma must agree with an equivalent one-k-point representation, and Eq. (79) must be satisfied.

9 GDF: auxiliary metrics and nuclear response

9.1 Fitting AO-pair densities

Gaussian density fitting represents AO-pair densities in an atom-centered auxiliary basis $\{P\}$. For a k-point pair $(\mathbf{k}_i, \mathbf{k}_j)$ with momentum transfer $\mathbf{q} = \mathbf{k}_j - \mathbf{k}_i$, the three-center source and two-center Coulomb metric are

$$B^{ij}_{P,\mu\nu} = (P\mathbf{q}|\mu\mathbf{k}_i, \nu\mathbf{k}_j)\,, \tag{97}$$

$$M^{\mathbf{q}}_{PQ} = (P\mathbf{q}|Q\mathbf{q})\,. \tag{98}$$

With the metric factorization

$$M^{\mathbf{q}} = L^{\mathbf{q}} L^{\mathbf{q}\dagger}, \tag{99}$$

the orthonormalized CDERI is

$$C^{ij} = L^{-1} B^{ij}. \tag{100}$$

A four-center integral is approximated by contracting two such CDERI factors.

9.2 The three-center source derivative

$B_{P,\mu\nu}$ contains an auxiliary center, a bra-AO center, and a ket-AO center. Its derivative for atom A is

$$\begin{aligned} B^{ij,Ax}_{P,\mu\nu} = {} & \delta_{A,A_P}\partial_{P_x} B^{ij}_{P,\mu\nu} + \delta_{A,A_\mu}\partial_{\mu_x} B^{ij}_{P,\mu\nu} \\ & + \delta_{A,A_\nu}\partial_{\nu_x} B^{ij}_{P,\mu\nu}. \end{aligned} \tag{101}$$

If an integral library's `ip1`/`ip2` operator differentiates an electronic coordinate, a translational identity may reverse the sign for a nuclear-center displacement. Each axis and atom slice must therefore be checked with a small component test.

A periodic GDF builder may combine short-range real-space, long-range reciprocal-space, and $\mathbf{q} = 0$ correction terms,

$$B^{ij} = B^{ij}_{\mathrm{SR}} + B^{ij}_{\mathrm{LR}} + \delta_{\mathbf{q},0} B^{ij}_{vbar}. \tag{102}$$

The derivative must use the same screening, lattice images, and reciprocal mesh as the central builder.

9.3 Metric and Cholesky derivatives

The two-center metric derivative contains the responses of both auxiliary centers,

$$M^{\mathbf{q},Ax}_{PQ} = \delta_{A,A_P}\partial_{P_x} M^{\mathbf{q}}_{PQ} + \delta_{A,A_Q}\partial_{Q_x} M^{\mathbf{q}}_{PQ}. \tag{103}$$

Differentiating Eq. (99) gives

$$M^{Ax} = L^{Ax} L^{\dagger} + L L^{Ax\dagger}. \tag{104}$$

Define $W = L^{-1} M^{Ax} L^{-\dagger}$. In the lower-Cholesky gauge,

$$L^{Ax} = L\,\Phi(W), \tag{105}$$

where Φ retains the strict lower triangle and half of the diagonal. The CDERI derivative is therefore

$$C^{Ax} = L^{-1} B^{Ax} - L^{-1} L^{Ax} C. \tag{106}$$

The second term is the metric/auxiliary response. Omitting it differentiates a functional form different from the one used for the fitted central energy.

9.4 J and K derivatives

Using the diagonal k-pair CDERI, define the Coulomb auxiliary density as

$$j_P = \frac{1}{N_k} \sum_{\mathbf{k}\mu\nu} C^{\mathbf{kk}}_{P,\mu\nu} D^{\mathbf{k}}_{\nu\mu}. \tag{107}$$

At fixed density,

$$j^{Ax}_P = \frac{1}{N_k} \sum_{\mathbf{k}\mu\nu} C^{\mathbf{kk},Ax}_{P,\mu\nu} D^{\mathbf{k}}_{\nu\mu}. \tag{108}$$

Consequently,

$$J^{\mathbf{k},Ax}_{\mu\nu} = \sum_P \left(j^{Ax}_P C^{\mathbf{kk}}_{P,\mu\nu} + j_P C^{\mathbf{kk},Ax}_{P,\mu\nu} \right). \tag{109}$$

Exchange similarly applies the product rule to each of its two CDERI factors. Reverse blocks of canonical k-pairs are reconstructed with the required complex conjugation and AO transpose.

9.5 Two appearances of auxiliary response in the gradient

First, C^{Ax} appears in the explicit target-Hartree derivative of Eq. (71). Second, GDF J/K derivatives enter the source HF CPHF source and Hessian action. Adding metric response only to the explicit target term while omitting it from the source response, or vice versa, breaks agreement with the full-energy finite difference.

9.6 The value-parity guard

Define the difference between the central CDERI reconstructed by the response builder and the CDERI stored by the actual SCF provider as

$$\Delta_C = \max |C_{\text{response builder}} - C_{\text{SCF storage}}|. \tag{110}$$

If Δ_C exceeds tolerance, even plausible B^{Ax} and M^{Ax} are not derivatives of the energy representation that was actually used, and the gradient must be rejected. Common causes are different values of ω, reciprocal meshes, auxiliary-basis ordering, k-pair orientation, or metric factorization gauges.

9.7 CPU/GPU boundaries

The CPU package generates central and derivative primitives with the PySCF periodic integral engine and performs linear algebra with NumPy/SciPy. The GPU package declares periodic two- and three-center primitive generation as a host boundary, while CDERI storage, Cholesky/metric response, dJ/dR, dK/dR, AX/Z-vector operations, and final contractions run on the CuPy device. `device="gpu"` does not claim that integral metadata are also generated on the GPU; it guarantees that supported numerical contractions do not silently call the CPU implementation.

GDF completion requires full-energy finite differences, the translation sum, and CDERI parity for distorted H_2O as well as H_2, for Gamma and non-Gamma k-meshes, and at both SZV and DZVP levels.

10 SCAN/r2SCAN meta-GGA energies and gradients

10.1 Semilocal ingredients

At each grid point, a spin-unpolarized MGGA evaluates

$$e_{\mathrm{xc}} = f(\rho, \sigma, \tau), \qquad \sigma = \nabla\rho \cdot \nabla\rho. \tag{111}$$

The weights entering the variational derivative are

$$v_\rho = \frac{\partial f}{\partial \rho}, \qquad v_\sigma = \frac{\partial f}{\partial \sigma}, \qquad v_\tau = \frac{\partial f}{\partial \tau}. \tag{112}$$

SCAN is a constraint-based functional designed to connect the uniform-electron gas, single-orbital, and slowly varying density limits. r2SCAN regularizes the numerically sensitive switching functions of SCAN and improves grid convergence, but it has the same derivative structure in terms of ρ, σ, and τ.

10.2 AO representation of τ

Writing Eq. (25) as an AO product gives

$$\tau(\mathbf{r}) = \frac{1}{2N_k} \sum_{\mathbf{k}\mu\nu} D^{\mathbf{k}}_{\mu\nu} \sum_{y=x,y,z} \partial_y \phi^*_{\nu\mathbf{k}}(\mathbf{r}) \partial_y \phi_{\mu\mathbf{k}}(\mathbf{r}). \tag{113}$$

The tau contribution to the target MGGA Fock matrix is

$$V^{\tau,\mathbf{k}}_{\mu\nu} = \frac{1}{2} \int_\Omega v_\tau(\mathbf{r}) \nabla\phi^*_{\nu\mathbf{k}}(\mathbf{r}) \cdot \nabla\phi_{\mu\mathbf{k}}(\mathbf{r}) \mathrm{d}\mathbf{r}. \tag{114}$$

The factor $1/2$ comes from Eq. (25). One must first determine whether the numerical integration library already incorporates the τ convention into its effective weight and then apply the factor exactly once.

10.3 Explicit nuclear derivative

At fixed density matrix,

$$E^{Ax}_{\mathrm{xc}} = \int_\Omega \left(v_\rho \rho^{Ax} + v_\sigma \sigma^{Ax} + v_\tau \tau^{Ax} \right) \mathrm{d}\mathbf{r}. \tag{115}$$

On a fixed-cell uniform grid, quadrature points and volume weights do not change with a nuclear coordinate, so only AO derivatives remain. The GGA part contains

$$\sigma^{Ax} = 2\nabla\rho \cdot \nabla\rho^{Ax}, \tag{116}$$

while the tau derivative is

$$\tau^{Ax}(\mathbf{r}) = \frac{1}{2N_k} \sum_{\mathbf{k}\mu\nu} D^{\mathbf{k}}_{\mu\nu} \sum_y \left[\partial_y \phi^{Ax*}_{\nu\mathbf{k}} \partial_y \phi_{\mu\mathbf{k}} + \partial_y \phi^*_{\nu\mathbf{k}} \partial_y \phi^{Ax}_{\mu\mathbf{k}} \right]. \tag{117}$$

Because an AO-center derivative and a spatial gradient appear together, second spatial derivatives of the AOs are required. This is the principal reason why an MGGA gradient kernel is more complicated than a GGA kernel.

10.4 Coupling to the nonvariational response

HF still defines the source orbitals in an MGGA target calculation. The Hessian A is therefore the same as in Eq. (33). The addition of Eq. (114) to the target Fock matrix changes the target VO gradient to

$$g_{ai}^{\mathbf{k},\mathrm{MGGA}} = \left[C_v^{\mathbf{k}\dagger}\left(h + J + V_\rho + V_\sigma + V_\tau\right)^{\mathbf{k}} C_o^{\mathbf{k}}\right]_{ai}. \tag{118}$$

After solving for the Z-vector, it is contracted with the source-HF nuclear perturbation through Eq. (76). Connecting the MGGA kernel only to the explicit XC term while retaining a PBE target-Fock VO right-hand side would give an incorrect nonvariational response.

10.5 Grid aliasing

High-frequency components of Gaussian AOs alias on a coarse grid. Since τ is a product of AO gradients, it is more sensitive to high reciprocal components than the density itself. MGGA calculations therefore require a separate convergence study of `cell.mesh`, or of the effective integration mesh. The resolved integer triplet and cutoff should be recorded together rather than changing one direction at a time.

The translational sum rule is a sensitive measure of grid aliasing. Even when the analytic–FD component error is small, a large

$$\left\| \sum_A \mathbf{g}_A^{\mathrm{xc}} \right\| \tag{119}$$

means that the explicit XC-grid derivative has not converged.

10.6 CPU/GPU parity and backend composition

CPU and GPU routes must use the same libxc functional label, density convention, mesh, and AO derivative order. Comparisons include not only total energy and final gradient, but also E_{xc}, the target-Fock VO block, the v_τ row derivative, the Z-vector right-hand side, and its residual. Completion cases for SCAN/r2SCAN include Gamma-point H_2, distorted H_2O, and sampled k-points.

The target MGGA and the source-HF density-fitting provider have separate responsibilities. The same ρ, $\nabla\rho$, and τ target path must therefore be tested with RSDF as well as FFTDF and GDF. An RSDF+MGGA case is a composition test in which the XC derivative of Eq. (115) and the source/metric auxiliary response of Eq. (130) operate simultaneously.

11 RSDF: range-separated source and metric response

11.1 Purpose of range separation

Range-separated density fitting (RSDF) uses the exact identity

$$\frac{1}{r} = \frac{\mathrm{erfc}(\omega r)}{r} + \frac{\mathrm{erf}(\omega r)}{r} \equiv v_{\mathrm{SR}}(r;\omega) + v_{\mathrm{LR}}(r;\omega). \tag{120}$$

The short-range part decays rapidly in a real-space lattice sum, whereas the long-range part is readily evaluated as a smooth reciprocal-space sum. The infinite sum is independent of ω, but under finite screening and on a finite mesh, ω is part of the numerical representation.

11.2 Source and metric parameters are independent

The exponent distributions optimized by the three-center source builder and the two-center metric builder differ, so PySCF RSDF may select different parameters for them. Write

$$B^{ij} = B^{ij}_{\mathrm{SR}}(\omega_B, \epsilon_B) + B^{ij}_{\mathrm{LR}}(\omega_B, \mathcal{G}_B) + \delta_{\mathbf{q},0} B^{ij}_{vbar}, \tag{121}$$

$$M^{\mathbf{q}} = M^{\mathbf{q}}_{\mathrm{SR}}(\omega_M, \epsilon_M) + M^{\mathbf{q}}_{\mathrm{LR}}(\omega_M, \mathcal{G}_M) + \delta_{\mathbf{q},0} M_0. \tag{122}$$

In general,

$$\omega_B \neq \omega_M, \qquad \mathcal{G}_B \neq \mathcal{G}_M. \tag{123}$$

Reusing the source parameter for the metric derivative is not a controlled approximation; it differentiates a different central CDERI.

11.3 Fixed-cell nuclear derivatives

Holding the cell, resolved source and metric meshes, ω_B, and ω_M fixed gives

$$B^{ij,Ax} = B^{ij,Ax}_{\mathrm{SR}} + B^{ij,Ax}_{\mathrm{LR}} + \delta_{\mathbf{q},0} B^{ij,Ax}_{vbar}, \tag{124}$$

$$M^{\mathbf{q},Ax} = M^{\mathbf{q},Ax}_{\mathrm{SR}} + M^{\mathbf{q},Ax}_{\mathrm{LR}} + \delta_{\mathbf{q},0} M^{Ax}_0. \tag{125}$$

The short-range derivative uses the same lattice-image list and screening precision as the central source or metric. The long-range derivative uses the same reciprocal vectors and structure factors. The integer mesh does not change under a nuclear derivative.

After Cholesky factorization of the resolved metric, Eq. (106) applies unchanged,

$$C^{Ax} = L^{-1} B^{Ax} - L^{-1} L^{Ax} C. \tag{126}$$

Both source and metric responses must be retained.

11.4 The $q = 0$ correction

$\mathbf{q} = 0$ does not mean Gamma-only. A diagonal pair $(\mathbf{k}_i, \mathbf{k}_i)$ in a sampled-k calculation also has $\mathbf{q} = 0$. Reciprocal equivalence is accounted for by a test such as

$$\delta_{\mathbf{q},0} = \begin{cases} 1, & \|\mathbf{k}_j - \mathbf{k}_i - \mathbf{G}\| < \epsilon_q \text{ for some reciprocal } \mathbf{G}, \\ 0, & \text{otherwise.} \end{cases} \tag{127}$$

The values and derivatives of source *vbar*, metric M_0, and the Ewald exchange correction must use the same pair orientation.

11.5 Canonical-pair storage

Memory can be reduced by storing only canonical (i, j) pairs. A reverse pair is reconstructed as

$$C^{ji}_{P,\nu\mu} = C^{ij*}_{P,\mu\nu}, \tag{128}$$

and the derivative must obey

$$C^{ji,Ax}_{P,\nu\mu} = C^{ij,Ax*}_{P,\mu\nu}. \tag{129}$$

Transposing the AO indices while omitting the complex conjugate may pass a Gamma test but produces imaginary leakage and gradient errors at non-Gamma k-points.

11.6 Where the RSDF response is used

As in GDF, the RSDF CDERI derivative enters the explicit target-Hartree derivative, the source-HF J/K nuclear derivative, and the CPHF source. The AX operation applies central RSDF J/K to a transition density. The complete structure can be summarized as

$$g_{Ax}^{\mathrm{RSDF}} = g_{Ax}^{1e} + g_{Ax}^{\mathrm{H}}[C, C^{Ax}] + g_{Ax}^{\mathrm{xc}} + g_{Ax}^{\mathrm{Pulay,t}} + g_{Ax}^{\mathrm{NN}} + z^{\mathrm{H}} B_{\mathrm{HF}}^{Ax}[C, C^{Ax}]. \tag{130}$$

11.7 The GPU host boundary

GPU4PySCF currently has no device integral engine that generates all periodic RSDF two- and three-center derivative primitives. Central and derivative integral metadata are therefore generated on the host and transferred to CuPy. Metric factorization, dC, dJ/dR, dK/dR, AX/Z-vector operations, and final assembly then run on the device. This boundary is recorded in the metadata and is distinct from a silent fallback to a CPU DC-DFT gradient kernel.

11.8 RSDF completion checks

RSDF validation proceeds in the following order:

1. Record the actual provider-resolved ω_B, source mesh, ω_M, and metric mesh.
2. Check value parity between the response builder's central CDERI and the SCF storage.
3. Compare source-only, metric-only, and total dC with component finite differences.
4. Test dJ/dR, dK/dR, the explicit target term, and the Z response separately.
5. Perform full-energy central finite differences at Gamma and sampled k-points.
6. Check CPU/GPU parity and the translational sum rule.

An empirical correction that becomes small only for a particular molecule when the auxiliary basis or source/metric mesh is changed is not accepted as completion evidence.

12 Numerical validation, convergence, and reporting units

12.1 Full-energy central finite differences

The authoritative numerical reference for an analytic gradient is

$$g_{Ax}^{\mathrm{FD}}(h) = \frac{E_{\mathrm{DC-DFA}}(\mathbf{R} + h\mathbf{e}_{Ax}) - E_{\mathrm{DC-DFA}}(\mathbf{R} - h\mathbf{e}_{Ax})}{2h}. \tag{131}$$

At each displaced geometry, the source HF problem is reconverged with the same tolerances and the target-DFA energy is evaluated anew. Holding the central density or MOs fixed while differentiating only an operator is a component diagnostic, not Eq. (131).

The leading truncation error is $O(h^2)$. If h is too small, SCF and floating-point noise are amplified as $O(\epsilon/h)$. One should locate a plateau over several steps h, $h/2$, and $h/4$, and verify that the analytic gradient agrees with that plateau.

12.2 Error norms

For a selected component set $\mathcal{I}$, record

$$\Delta_{\max} = \max_{(A,x)\in\mathcal{I}} |g_{Ax} - g_{Ax}^{\mathrm{FD}}|, \tag{132}$$

$$\Delta_{\mathrm{RMS}} = \sqrt{\frac{1}{|\mathcal{I}|} \sum_{(A,x)\in\mathcal{I}} |g_{Ax} - g_{Ax}^{\mathrm{FD}}|^2}. \tag{133}$$

A calculation that agrees for only one axis or one atom is not a full-gradient validation. A completion gate includes all $3N_{\mathrm{atom}}$ components of a distorted geometry.

12.3 SCF and Z-vector residuals

The energy and density tolerances of each displaced HF calculation must be tighter than the target gradient accuracy. The HF orbital residual, density change, and energy change are recorded together. For the Z-vector, record the maximum residual of Eq. (40) and a condition estimate. Even if CPU and GPU produce the same gradient, parity is not established if both rely on cancellation with a large residual.

12.4 Convergence of the representation

After analytic–FD agreement is established, converge one numerical parameter at a time:

- primary Gaussian basis and GTH pseudopotential,
- FFTDF cutoff and resolved integer mesh,
- target-XC integration mesh,
- GDF/RSDF auxiliary basis,
- GDF screening and reciprocal mesh,
- RSDF source ω_B/mesh and metric ω_M/mesh, and
- k-point mesh.

Do not report a cutoff without its resolved mesh. The same cutoff gives a different integer mesh when the lattice length changes. RSDF metadata must not collapse source and metric parameters into a single `omega` or `mesh`.

12.5 Symmetry-independent checks

Define the translational sum-rule error as

$$\Delta_{\mathrm{trans}} = \left\| \sum_A \mathbf{g}_A \right\|_2. \tag{134}$$

With only symmetric H_2, missing terms can cancel by symmetry. A mandatory test therefore uses H_2O with slightly distorted bonds, angle, and overall orientation. Both Gamma and sampled k-points, and both real and complex tensor paths, are included.

Equivalent Gamma representations must satisfy

$$E_\Gamma = E_{N_k=1}, \qquad \mathbf{g}_\Gamma = \mathbf{g}_{N_k=1}. \tag{135}$$

This catches shape-adapter errors that incorrectly remove or retain a singleton k-axis.

12.6 CPU/GPU parity

For identical cells, bases, pseudopotentials, meshes, k-points, and SCF tolerances, record

$$\Delta_E^{\mathrm{CPU/GPU}} = |E_{\mathrm{CPU}} - E_{\mathrm{GPU}}|, \tag{136}$$

$$\Delta_g^{\mathrm{CPU/GPU}} = \max_{Ax} |g_{Ax}^{\mathrm{CPU}} - g_{Ax}^{\mathrm{GPU}}|. \tag{137}$$

GDF/RSDF integral generation may be an explicit GPU host boundary. A GPU path that simply calls the CPU package's final gradient and returns that value is not parity; it is fallback. Ownership is verified through capability and timing metadata.

12.7 Initial acceptance thresholds

The initial completion targets for this project are

Test	Threshold
Analytic vs converged full-energy FD	$\Delta_{\max} \leq 10^{-5}$ Ha/Bohr
CPU/GPU energy	$\Delta_E \leq 10^{-8}$ Ha
CPU/GPU gradient	$\Delta_g \leq 10^{-7}$ Ha/Bohr
Translation norm	$\Delta_{\mathrm{trans}} \leq 10^{-6}$ Ha/Bohr

Thresholds are interpreted together with basis, cutoff, and FD-step convergence evidence. Exact analytic–FD agreement on a coarse mesh does not imply that the resulting gradient is the converged physical gradient.

12.8 Result metadata

An energy/gradient result preserves at least the following information in a machine-readable form:

- method, target XC, source reference, charge, and spin,
- atomic coordinates, lattice, and the unit of each,
- basis and pseudopotential,
- Gamma/k-point list and weights,
- DF backend and auxiliary basis,
- cutoff and resolved FFT/XC meshes,
- RSDF source and metric settings,
- HF convergence and energy timings,
- Z dimension, residual, condition estimate, and gradient-component timings,
- device, host boundaries, and a silent-fallback flag, and
- gradient-array unit and the force-sign convention.

The canonical gradient output is a NumPy array in Ha/Bohr. Rather than duplicate large arrays in JSON, metadata may record the shape, dtype, unit, checksum, and path to an NPY file.

12.9 Common failure signatures

Observation	First causes to inspect
Gamma passes but k-points fail	$1/N_k$, conjugation, reverse pairs, $q = 0$
GDF error grows from small to larger systems	metric/auxiliary-center response
Error changes discontinuously with cutoff	different resolved meshes at displaced geometries
Only MGGA has a large net translation	τ factor, second AO derivatives, grid aliasing
RSDF value parity fails	mixed source/metric ω or mesh
Z residual is small but full FD fails	explicit hcore/J/XC/Pulay derivative
Only CPU/GPU totals agree, components differ	silent fallback or different ownership
Stable multigrid energy and gradient but incorrect FD	different target mesh/tasks or double-counted J owner
Illegal address at a large target mesh without OOM	index/buffer violation in an earlier asynchronous custom kernel

13 Independent ownership of target multigrid and source DF

13.1 Two independent choices

A periodic DC-DFT calculation contains two numerical choices with different roles. The *source two-electron backend* evaluates Coulomb, exchange, and orbital-Hessian actions for the source HF problem. This backend is FFTDF, GDF, or RSDF. The *target integrator* evaluates target J and E_{xc} at the converged HF density. Calling both choices one "DF method" obscures their separate responsibilities.

We denote a route by the ordered pair

$$\mathcal{R} = (\mathcal{B}_{\mathrm{source}}, \mathcal{I}_{\mathrm{target}}), \qquad \mathcal{B}_{\mathrm{source}} \in \{\mathrm{FFTDF}, \mathrm{GDF}, \mathrm{RSDF}\}. \tag{138}$$

Changing the target integrator must not change the source HF density D^{HF}, HF energy, occupied and virtual orbitals, or source Hessian A^{HF}. It changes the target quadrature in $E_{\mathrm{DFA}}[D^{\mathrm{HF}}]$, the target Fock matrix, the orbital-gradient right-hand side, and the explicit target derivative.

13.2 Three target-integration modes

The current GPU route distinguishes three modes.

Mode	Target J	Target XC	Source HF/AX
`uniform`	selected DF	uniform KNumInt	selected DF
`multigrid_xc`	selected DF	multigrid	selected DF
`multigrid_j_xc`	multigrid	multigrid	selected DF

For `multigrid_xc`, called M1, the target effective potential is

$$V_{\mathrm{eff}}^{\mathrm{M1}}[D] = J_{\mathcal{B}_{\mathrm{source}}}[D] + V_{\mathrm{xc}}^{\mathrm{MG}}[D]. \tag{139}$$

M1 keeps the finite-auxiliary GDF/RSDF representation of target J and replaces only the XC quadrature.

For `multigrid_j_xc`, called M2,

$$V_{\mathrm{eff}}^{\mathrm{M2}}[D] = J_{\mathrm{MG}}[D] + V_{\mathrm{xc}}^{\mathrm{MG}}[D]. \tag{140}$$

The multigrid J_{MG} and a finite-auxiliary GDF/RSDF J are different finite numerical representations. They approach the same Coulomb limit when their meshes and auxiliary bases are separately converged, but their finite-setting energies need not agree. M2 is therefore a separate target model, not merely a drop-in acceleration of GDF or RSDF.

13.3 Ownership in the energy and gradient

The target Coulomb derivative must be included exactly once. The ownership rule is

$$\mathbf{g}_{\mathrm{target}} = \begin{cases} \mathbf{g}_J^{\mathcal{B}_{\mathrm{source}}} + \mathbf{g}_{\mathrm{xc}}^{\mathrm{uniform}}, & \mathcal{I} = \texttt{uniform}, \\ \mathbf{g}_J^{\mathcal{B}_{\mathrm{source}}} + \mathbf{g}_{\mathrm{xc}}^{\mathrm{MG}}, & \mathcal{I} = \texttt{multigrid_xc}, \\ \mathbf{g}_{J+\mathrm{xc}}^{\mathrm{MG}}, & \mathcal{I} = \texttt{multigrid_j_xc}. \end{cases} \tag{141}$$

Adding the selected-DF target dJ/dR in the last line double counts the Coulomb response. Omitting it in M1 differentiates a Hamiltonian different from the one used for the energy.

The target hcore and GTH pseudopotential remain owned by the selected one-electron provider in all three modes. Attaching a multigrid object to the entire target KS object can implicitly redirect upstream `get_hcore()` or `get_j()` dispatch. The implementation must instead combine the low-level target XC/J outputs explicitly and report separate hcore, PP, J, and XC owners.

13.4 Why target multigrid does not replace the HF Hessian

The DC-DFT adjoint equation differentiates the source HF stationarity constraint. In

$$A_{\mathrm{HF}}^{T} z = -g_{\mathrm{target}}, \tag{142}$$

the right-hand side depends on the target integrator, whereas A_{HF} remains the source Hessian. Replacing AX J/K by multigrid after selecting M1 or M2 would solve the response of a different stationarity constraint than the one satisfied by the source density.

The workflow is therefore:

1. converge source HF with GDF or RSDF and obtain D^{HF};
2. build $E_{\mathrm{DFA}}[D^{\mathrm{HF}}]$ and F_{target} with the selected target integrator;
3. form g_{target} from $F_{\mathrm{target},vo}$;
4. solve for z with source GDF/RSDF AX;
5. assemble target explicit and source-HF response derivatives.

13.5 Source and target meshes

The `cell.mesh` resolved from `cell.ke_cutoff` may be used by FFT-based source operators, the GTH local potential, and other reciprocal operations. The target multigrid mesh is an independent parameter for target XC/J quadrature. Equal default values do not imply equal accuracy meaning.

Convergence should be separated as

$$\lim_{\mathcal{M}_s\to\infty}\lim_{\mathcal{M}_t\to\infty}\mathbf{g}(\mathcal{M}_s,\mathcal{M}_t). \tag{143}$$

Reporting only one cutoff does not identify which representation is converged. `target_multigrid_mesh=null` inherits the current cell mesh; it is not an automatic memory cap. Large-cell reports must record the resolved triplet and number of grid points.

Multigrid distributes work over Gaussian/locality levels, but it remains a finite quadrature. It is not automatically more accurate or more translation invariant than a dense uniform grid. Energy phase scans, translation sums, and complete-gradient mesh convergence test distinct properties and should all be reported.

13.6 Workspace reuse and lifetime

Multigrid tasks, sorted Gaussian pairs, and the target Fock built during the target energy can be reused by the gradient. Using a different multigrid object or mesh in the gradient does not differentiate the central energy. Conversely, keeping the large task cache after its final use overlaps its memory with later GDF/RSDF response work.

The safe lifetime is

$$\text{build at target energy} \longrightarrow \text{reuse at target gradient} \longrightarrow \text{release before DF response.} \tag{144}$$

This is ownership management after the last consumer, not an approximation that discards grid points.

13.7 Recommended scope

M1 is the conservative opt-in route because it retains source HF and the finite-auxiliary target J while replacing only target XC. M2 can be analytically consistent, but it also changes the target-J representation and requires separate convergence. A PP-only multigrid route is inconsistent until central hcore, explicit PP derivatives, the nuclear CPHF source, and the Z contraction all use the same representation.

14 Matrix-free Z-vectors and bounded-memory response

14.1 Response dimension and dense-matrix cost

At k-point $\mathbf{k}$, occupied–virtual rotations have complex dimension $n_o^{\mathbf{k}}n_v^{\mathbf{k}}$. After the realification of Eq. (38), the total real dimension is

$$m = 2\sum_{\mathbf{k}} n_o^{\mathbf{k}}n_v^{\mathbf{k}}. \tag{145}$$

A dense real Hessian is $m \times m$ and its float64 storage alone is

$$M_{\text{dense}} = 8m^2 \text{ bytes}. \tag{146}$$

Factorization and condition diagnostics require additional copies and workspace. The dense route is useful for equation-level parity on small reference systems, but it is not a production scaling strategy.

Each dense column requires one AX application to a basis vector e_j. This does not mean that m^2 separate equations are solved; rather, m operator actions are collected to materialize m^2 matrix elements.

14.2 Matrix-free Hessian actions

The matrix-free route never stores A. For a trial vector u, it evaluates

$$u \longmapsto D^{(1)}(u) \longmapsto \delta J[D^{(1)}] - \tfrac{1}{2}\delta K[D^{(1)}] \longmapsto Au. \tag{147}$$

The first map constructs a Hermitian transition density from VO rotations. The second applies the selected FFTDF/GDF/RSDF central J/K provider. The last projects the AO result back to VO blocks and adds the orbital-energy gap.

If a Krylov solve takes 21 iterations, it does not select only 21 important components out of the m unknowns. Every iteration applies AX to a complete length-m trial vector; 21 Krylov directions were sufficient to represent the solution to the requested residual.

14.3 MINRES and the energy-gap preconditioner

The realified HF Hessian is ideally real symmetric, but need not be positive definite. MINRES is a Lanczos/Krylov method that reduces the residual norm for symmetric indefinite systems. Starting from $z_0 = 0$, the order-p subspace is

$$\mathcal{K}_p(A, b) = \mathrm{span}\{b, Ab, A^2b, \ldots, A^{p-1}b\}. \tag{148}$$

A simple preconditioner is the independent-particle gap,

$$P_{ai\mathbf{k},ai\mathbf{k}} = \varepsilon_a^{\mathbf{k}} - \varepsilon_i^{\mathbf{k}}, \qquad P^{-1}r \approx A^{-1}r. \tag{149}$$

This does not approximate the final physical solution by dropping two-electron coupling. MINRES still applies the full A; P^{-1} rescales poorly conditioned directions so the Krylov subspace converges faster. Positive VO gaps, dtype, and k-point ordering are checked before iteration.

Convergence is accepted using the explicitly recomputed equation residual

$$\eta_z = \frac{\|A_R^T z_R + g_R\|_\infty}{\max(1, \|g_R\|_\infty)}. \tag{150}$$

If needed, iterative refinement solves

$$A_R^T \delta z_R = -\left(A_R^T z_R + g_R\right), \qquad z_R \leftarrow z_R + \delta z_R \tag{151}$$

with the same matrix-free operator.

14.4 Symmetry, complex k-points, and solver selection

MINRES assumes that the realified operator is symmetric within numerical tolerance. Incorrect reverse-pair conjugation, $1/N_k$, occupation factors, or J/K transposes increase $\|A_R - A_R^T\|$ in a dense test and can destabilize the matrix-free solve. Small Gamma and sampled-k systems must establish dense/matrix-free parity for solutions, gradients, and residuals.

An automatic policy may use dense reference solves at small m and matrix-free solves at large m. Rebuilding a large dense matrix merely to obtain an exact condition number defeats the purpose. Production reports should state that the condition number was not evaluated and preserve the iteration count, residual, and minimum gap instead.

14.5 Batching nuclear sources

The Z-vector is solved once, but B^{Ax} contains $3N_A$ perturbations. Materializing all AO/VO response tensors together scales approximately as

$$M_B \sim 3N_A N_k N_{\mathrm{AO}}^2 \, s_{\mathrm{dtype}}. \tag{152}$$

For a component batch $\mathcal{C}$, one evaluates

$$g_{Ax}^{\mathrm{resp}} = z^T B^{Ax}, \qquad (A,x) \in \mathcal{C}, \tag{153}$$

and keeps only the scalar gradient. The batch size is determined from free memory, dtype, AO dimension, and temporary count. An OOM retry starts from the same component with a smaller batch and must not add completed components twice.

14.6 Adjoint contractions instead of full derivative tensors

Let a whitened three-center factor be $C = L^{-1}B$. Its nuclear derivative is

$$C^{Ax} = L^{-1}\left(B^{Ax} - L^{Ax}C\right), \tag{154}$$

but the gradient usually needs only

$$\mathcal{G}^{Ax} = \mathrm{Re}\langle W_C, C^{Ax}\rangle. \tag{155}$$

Defining $W_B = L^{-\dagger}W_C$ gives

$$\mathcal{G}^{Ax} = \mathrm{Re}\langle W_B, B^{Ax}\rangle - \mathrm{Re}\langle W_B, L^{Ax}C\rangle. \tag{156}$$

Source and metric derivative blocks can therefore be accumulated directly into scalars without storing a full $(3N_A, N_{\mathrm{aux}}, N_{\mathrm{AO}}^2)$ tensor. This is reverse-mode or adjoint algebra, not a screening approximation.

14.7 Signed two-density polarization

Write the fixed-density restricted-HF two-electron derivative as

$$Q^{Ax}(X) = \frac{1}{2N_k}\sum_{\mathbf{k}} \mathrm{Re}\,\mathrm{Tr}\left[X^{\mathbf{k}} L^{Ax}[X]^{\mathbf{k}}\right]. \tag{157}$$

Because L^{Ax} is linear in density, the response between reference density D and Z transition density T obeys the exact polarization identity

$$R_{2e}^{Z,Ax} = Q^{Ax}(D+T) - Q^{Ax}(D-T). \tag{158}$$

The equivalent direct form is

$$R_{2e}^{Z,Ax} = \frac{1}{N_k}\sum_{\mathbf{k}} \mathrm{Re}\left\{\mathrm{Tr}[D^{\mathbf{k}} L^{Ax}[T]^{\mathbf{k}}] + \mathrm{Tr}[T^{\mathbf{k}} L^{Ax}[D]^{\mathbf{k}}]\right\}. \tag{159}$$

Processing the signed states in one SR/LR/metric traversal can reuse grids and integrals that two independent quadratic traversals would read twice. Both Coulomb and exchange orientations, auxiliary-metric response, the Ewald exchange correction, and k-pair conjugation must be retained. Evaluating only $D+T$ mixes self terms and is not Eq. (159).

14.8 Streaming, block size, and object lifetime

If a reciprocal derivative treats b_G vectors and all AO pairs together, the main working set scales as

$$M_{\mathrm{block}} \sim b_G N_k N_{AO}^2 s_{\mathrm{complex}}\, n_{\mathrm{temporary}}. \tag{160}$$

Reducing b_G lowers peak memory while retaining every reciprocal vector. Smaller blocks may add kernel launches, so a free-memory fraction and a hard cap are used together and the resolved block size and count are reported.

CDERI, multigrid tasks, and response temporaries should also be released after their last consumer. The central distinction is

$$\underbrace{\text{batching, streaming, adjoint, lifetime}}_{\text{same finite representation}} \neq \underbrace{\text{looser screening, smaller mesh, smaller basis}}_{\text{changed representation}}. \tag{161}$$

The first group must agree within roundoff and is tested by parity. The second changes both cost and accuracy and requires a convergence study.

14.9 Algorithmic references

MINRES was introduced by C. C. Paige and M. A. Saunders, SIAM J. Numer. Anal. 12, 617–629 (1975). This project combines that general algorithm with an HF Hessian action and orbital-gap preconditioner. The PySCF design and periodic Gaussian framework are described by Sun et al., WIREs Comput. Mol. Sci. 8, e1340 (2018).

15 Worked computational workflow and result interpretation

This chapter connects the preceding equations to the order of operations in a restricted closed-shell Gamma calculation. It is not a molecule-specific input tutorial; it follows which object is built at each step and which numerical representation owns it.

15.1 Step 0: define the numerical problem

Consider an RSDF source backend, the `multigrid_xc` target integrator, and r2SCAN. Before the calculation, fix

- coordinates, lattice, units, charge, and spin;
- primary basis, GTH pseudopotential, and auxiliary basis;
- Gamma or an explicit k-point list and weights;
- source cutoff/mesh and RSDF source/metric ω and meshes;
- target multigrid mesh; and
- HF, gradient, and Z-vector tolerances.

Recording both source and target meshes defines a reproducible problem more completely than saying only that the cutoff was 500 Ha.

15.2 Step 1: source HF

RSDF builds $J[D]$ and $K[D]$ in the iteration

$$F_{\mathrm{HF}} = h + J[D] - \tfrac{1}{2}K[D]. \tag{162}$$

After convergence, retain D^{HF}, MO coefficients, occupations, orbital energies, and the source DF provider. There is no r2SCAN target optimization in this step. If switching the target integrator from uniform to multigrid changes this source state, the ownership implementation is incorrect.

15.3 Step 2: fixed-density target energy

Evaluate the target functional at the same D^{HF}. In M1,

$$E_{\mathrm{target}} = \mathrm{Tr}[D^{\mathrm{HF}}h] + E_J^{\mathrm{RSDF}}[D^{\mathrm{HF}}] + E_{\mathrm{xc}}^{\mathrm{MG}}[D^{\mathrm{HF}}] + E_{NN}, \tag{163}$$

and

$$F_{\mathrm{target}} = h + J_{\mathrm{RSDF}}[D^{\mathrm{HF}}] + V_{\mathrm{xc}}^{\mathrm{MG}}[D^{\mathrm{HF}}]. \tag{164}$$

No SCF is run with this Fock matrix. The energy workspace retains the target Fock and multigrid tasks for reuse by the gradient.

15.4 Step 3: nonstationary right-hand side

Project the target Fock into the source HF VO block,

$$g_{ai} = 2(C_v^\dagger F_{\mathrm{target}} C_o)_{ai}. \tag{165}$$

If $g = 0$, the target energy happens to be stationary in the HF orbitals and there is no orbital-response contribution. In a general DC-DFT calculation, $g \neq 0$ and becomes the Z-vector right-hand side.

15.5 Step 4: source HF Z-vector

For small real dimension m, a dense A_R provides a reference solution. Large calculations use matrix-free AX with RSDF transition-density J/K and MINRES. The report should contain

- dense or matrix-free solver selection;
- real response dimension and iteration count;
- minimum orbital gap and preconditioner;
- final scaled residual η_z; and
- total and mean-iteration solver wall times.

A small iteration count alone is not an accuracy certificate. The explicit residual and small-system dense parity are the completion evidence.

15.6 Step 5: explicit target gradient

At fixed target density, evaluate

- selected one-electron h^{Ax} and GTH PP derivatives;
- the RSDF target dJ/dR;
- the multigrid target dE_{xc}/dR;
- the target energy-weighted-density contraction with dS/dR; and
- the nuclear–nuclear derivative.

Because this is M1, RSDF owns target dJ/dR and multigrid contributes XC only. In M2, combined multigrid $dJ + dE_{\mathrm{xc}}$ is used and the RSDF target-dJ route is disabled.

15.7 Step 6: source response gradient

For each Ax, build the one-electron, overlap, and source-RSDF two-electron nuclear source B^{Ax} and accumulate $z^T B^{Ax}$. Neither full B nor full C^{Ax} needs to remain resident. Nuclear-component batches, source/metric adjoint contractions, and reciprocal streaming leave only an $(N_A, 3)$ scalar array.

15.8 Step 7: final assembly

Equation (78) gives

$$\mathbf{g} = \mathbf{g}_{\mathrm{explicit,target}} + \mathbf{g}_{\mathrm{Pulay}} + \mathbf{g}_{\mathrm{source,response}} + \mathbf{g}_{NN}. \tag{166}$$

The public output is a gradient, not a force, and is reported in Ha/Bohr.

15.9 Shape and ownership trace

Object	Representative shape	Owner	Next consumer
D^{HF}	(N_k, N_{AO}, N_{AO})	source HF	target energy, AX
F_{target}	same	target integrator + DF	VO right-hand side
g_{vo}	per-k (N_v, N_o)	common orbital layer	Z solver
z_{vo}	per-k (N_v, N_o)	source HF Hessian	nuclear response
Target XC gradient	$(N_A, 3)$	multigrid	final assembly
RSDF response	$(N_A, 3)$	RSDF adjoint	final assembly
Final gradient	$(N_A, 3)$	common assembly	report/FD

15.10 Validation order

A completed large calculation is not sufficient validation. A useful order is

1. primitive matrix/response parity on a small distorted geometry;
2. Gamma versus explicit one-k-point shape parity;
3. sampled-k conjugation and $1/N_k$ checks;

4. selected Cartesian full-energy finite differences;

5. a complete $3N_A$ small-system FD gate;

6. separate convergence of source mesh, target mesh, auxiliary basis, and k-mesh; and

7. larger-system timing and memory measurements.

Analytic–FD agreement establishes that a chosen finite representation was differentiated correctly. Mesh convergence separately establishes that the finite representation is close enough to the desired physical limit.

15.11 Reading timing and memory reports

Decompose end-to-end time into source HF, target energy, target explicit gradient, Z-vector, one-electron/PP, DF response, and final reporting. High CPU use in a GPU calculation may reflect host integral generation, task construction, or packing and does not by itself imply a final CPU fallback. Use ownership metadata together with stage timing.

Peak VRAM must also be assigned to a stage. OOM is an allocation failure, whereas `CUDA_ERROR_ILLEGAL_ADDRESS` is an invalid access by a launched kernel. CUDA is asynchronous, so the next CuPy operation in a traceback may not be the first failing operation. A large-grid error should be localized with blocking diagnostics before it is hidden by lowering a cutoff; the physical representation and a kernel limitation are separate issues.

15.12 Further reading

The periodic Gaussian software framework is described in the PySCF framework paper. The range-separated three-center construction used by RSDF is described by Ye and Berkelbach, J. Chem. Phys. 154, 131104 (2021). The SCAN and r2SCAN definitions are given in Phys. Rev. Lett. 115, 036402 (2015) and J. Phys. Chem. Lett. 11, 8208–8215 (2020), respectively. Implementation-specific ownership and validation mappings are maintained in `docs/implementation/Equation_Code_Map.md`.

A Tensor shapes, complex conventions, and the equation map

A.1 Canonical tensor shapes

The following table defines the canonical shapes used by the implementation guide and source docstrings.

Quantity	Shape	Convention
$D^{\mathbf{k}}$	$(N_k, N_{\mathrm{AO}}, N_{\mathrm{AO}})$	includes occupation factor 2
$F_{\mathrm{t}}^{\mathbf{k}}$	$(N_k, N_{\mathrm{AO}}, N_{\mathrm{AO}})$	Hermitian
$F_{vo}^{\mathbf{k}}$	per-k (N_v, N_o)	target Z right-hand side
$z^{\mathbf{k}}$	per-k (N_v, N_o)	complex; realified solver
$h^{Ax,\mathbf{k}}$	$(N_A, 3, N_k, N_{\mathrm{AO}}, N_{\mathrm{AO}})$	full AO-matrix derivative
$S^{Ax,\mathbf{k}}$	same	full overlap derivative
$C_{P\mu\nu}^{ij}$	per-pair $(N_{\mathrm{aux}}, N_{\mathrm{AO}}, N_{\mathrm{AO}})$	canonical k-pair
$C_{P\mu\nu}^{ij,Ax}$	atom/axis + CDERI shape	source + metric response
g_{Ax}	$(N_A, 3)$	Ha/Bohr; not a force

At Gamma, the singleton k-axis is retained internally and restored to the public PySCF shape only at the API boundary. This prevents Gamma and k-point implementations from silently adopting different equations.

A.2 Complex-to-real Hessian blocks

For a complex response $U = X + iY$, suppose the real-linear map has the form

$$\delta F(U) = \mathcal{A}U + \mathcal{B}U^*. \tag{167}$$

Its realified operator is

$$\begin{pmatrix} \mathrm{Re}\,\delta F \\ \mathrm{Im}\,\delta F \end{pmatrix} = \begin{pmatrix} \mathrm{Re}(\mathcal{A}+\mathcal{B}) & -\mathrm{Im}(\mathcal{A}-\mathcal{B}) \\ \mathrm{Im}(\mathcal{A}+\mathcal{B}) & \mathrm{Re}(\mathcal{A}-\mathcal{B}) \end{pmatrix} \begin{pmatrix} X \\ Y \end{pmatrix}. \tag{168}$$

Production code need not materialize the full four-block matrix; it can supply the same map through matrix-vector products.

A.3 Summary of the Cholesky-response gauge

For $M = LL^\dagger$ and $W = L^{-1}M^{Ax}L^{-\dagger}$, define

$$\Phi(W)_{ij} = \begin{cases} W_{ij}, & i > j, \\ \frac{1}{2}W_{ii}, & i = j, \\ 0, & i < j. \end{cases} \tag{169}$$

Then $L^{Ax} = L\Phi(W)$. If the metric is complex Hermitian, diagonal imaginary noise may be removed within tolerance, but a large anti-Hermitian residual is an error.

A.4 A $G = 0$ and $q = 0$ checklist

1. The neutral Hartree $G = 0$ convention must be identical in the central energy and derivative.
2. Check the constant local-GTH term together with the ionic Ewald charge.
3. Keep the GDF/RSDF source *vbar* and metric correction separate.
4. Diagonal pairs in sampled-k calculations also have $q = 0$.
5. Include the overlap derivative of the exchange `exxdiv='ewald'` correction.
6. Do not use Gamma-only success as evidence for general $q = 0$ handling.

A.5 Principles for linking equations to code

Actual function names and validation tests are maintained in `docs/implementation/Equation_Code_Map.md`. Each mapping records an equation label, public or private function, input/output shapes, dtype, device, unit, and validation test. If an equation and source implementation diverge, the map and test are updated in the same change rather than modifying only one side.

The principal links are

Equation	Physical implementation owner
(19)	fixed-density target energy

(35)	common CPU/GPU adjoint response
(78)	backend-independent gradient assembly
(87)	GTH nonlocal-projector response
(92)	FFTDF J derivative
(106)	GDF source/metric response
(117)	SCAN/r2SCAN explicit XC derivative
(126)	RSDF independent source/metric response
(141)	target J/XC ownership and multigrid dispatch
(147)	matrix-free source-HF Hessian action
(156)	bounded-memory GDF/RSDF adjoint response

B References and reproducibility notes

PySCF and GPU4PySCF source files and examples are the primary references for API and tensor conventions. CP2K source and data provide comparison points for GTH pseudopotentials, periodic $G = 0$ treatment, cutoff/grid behavior, and force formulations; their GPW and auxiliary-density algorithms must not be identified with the PySCF GDF/RSDF equations. When external source code is consulted, its license and attribution requirements are checked and the private-API version is retained in result metadata.